CONVOLUTIONAL NEURAL NETWORK ACCELERATOR USING VERILOG HDL

Y. Naga Lokesh Manikanta (2411EC010102) | V. Narendra Varma (2411EC010101)

Project Guide: Mrs. C. Sheeba Thangapushpam, Associate Professor | Department of ECE, Malla Reddy University

1. Problem & Motivation

Executing Convolutional Neural Networks (CNNs) on software CPUs/ GPUs incurs high power dissipation (>100W) and latency bottlenecks due to sequential instruction fetching.

Solution: Custom RTL hardware accelerators on Field Programmable Gate Arrays (FPGAs) leverage spatial parallelism and deterministic execution to deliver low-power Edge AI.

2. Hardware Objectives

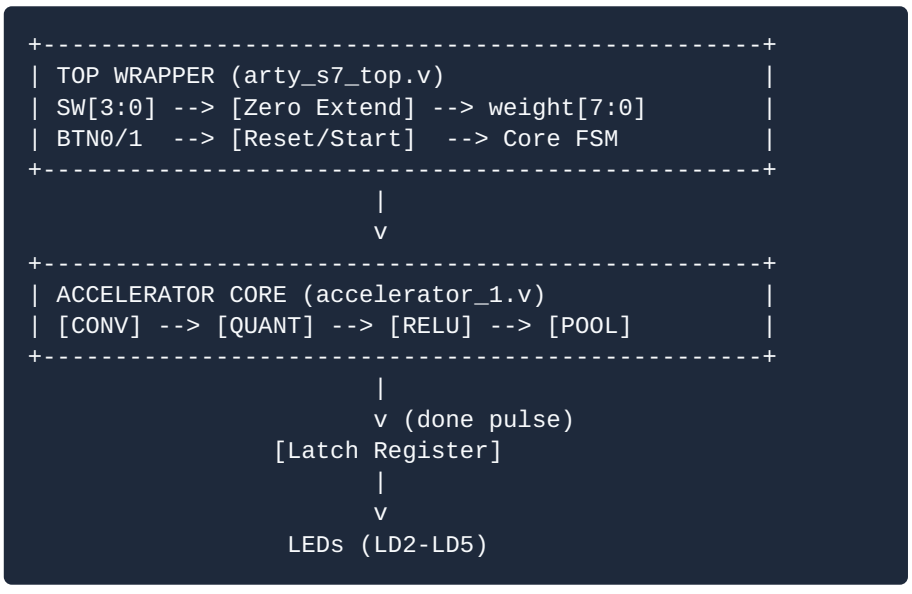
- Design a 4-stage pipeline core in Verilog HDL comprising Convolution, Quantization, ReLU, and Pooling primitives.
- Construct an FSM-driven Top Wrapper (`arty_s7_top.v`) to bind physical FPGA pins to internal logic.
- Achieve exact fixed-point scale alignment using arithmetic bit-shifting.
- Deploy and verify on Digilent Arty S7-25 FPGA hardware.

3. Data Pipeline & Formulae

Fixed-point arithmetic alignment across stages:

- **Stage 1 (Conv):** `conv_out = weight × act_in` (16-bit)
- **Stage 2 (Quant):** `quant_out = conv_out >> 4` (Divide by 16)
- **Stage 3 (ReLU):** `relu_out = max(0, quant_out)`
- **Stage 4 (Pool):** Asserts 1-cycle done pulse

4. Hardware System Block Diagram



5. Accelerator Core FSM (Verilog)

```
// FSM State Transition
always @(posedge clk or posedge reset) begin
  if (reset) begin
    state <= IDLE; final_out <= 8'sd0; done <= 1'b0;
  end else begin
    case (state)
      IDLE:  if (start) state <= CONV;
      CONV:  state <= QUANT;
      QUANT: state <= RELU;
      RELU:  state <= POOL;
      POOL:  begin
               if (pool_valid) begin
                 final_out <= pool_out;
                 done <= 1'b1; state <= IDLE;
               end
             end
    endcase
  end
end
```

6. Vivado Simulation Verification

Verified in Vivado Behavioral Simulator (100 MHz clock):

- **Vector 1:** Weight=6, Act=4 → Conv=24 → Final=1
- **Vector 2:** Weight=10, Act=2 → Conv=20 → Final=1
- **Board Test:** SW=3 (8'h03), Act=16 → Final=3

7. Spartan-7 Resource Utilization

Synthesis targeting AMD/Xilinx XC7S25-CSGA324:

28 (0.19%) SLICE LUTs	**34 (0.12%)** SLICE FLIP-FLOPS
1 (1.25%) DSP48E1 BLOCKS	**11 (7.33%)** IO PINS

8. Hardware Testing & Conclusion

Physical Board Test: Toggling SW0 and SW1 UP (Value = 3) on the Arty S7 board and pressing BTN1 illuminated LD2 and LD3 (Binary 0011), proving 1-to-1 scale alignment on silicon.

Conclusion: The core uses <0.2% FPGA resources, demonstrating a lightweight, deterministic IP core for Edge AI.